

# PROBLEM SET 1

## 3. Presentation

Consider the following characterization (in the steady state) of the Mexican labor market:

$$(f_{UI} + f_{UF} + f_{UO})U = If_{IU} + Ff_{FU} + Of_{OU}, \quad (1)$$

$$(f_{IU} + f_{IO} + f_{IF})I = Uf_{UI} + Of_{OI} + Ff_{FI}, \quad (2)$$

$$(f_{FU} + f_{FO} + f_{FI})F = Uf_{UF} + Of_{OF} + If_{IF}, \quad (3)$$

$$I + F + U + O = k, \quad (4)$$

where  $I$ ,  $F$ ,  $U$ , and  $O$  denote the number of informally employed, formally employed, unemployed, and individuals out of the labor force, respectively. Moreover,  $f_{AB}$  denotes the labor market gross flow from state  $A$  to state  $B$ , and  $k$  is the (fixed) size of the working-age population. Gross flows calculated by Leyva and Urrutia (2023) for Mexico are available in the companion dataset `labor_market_flows.xlsx`, Sheet `flows4`.

The main objective of this analysis is to quantify the contribution of each of the twelve labor market flows to the variability of the informal employment-to-population ratio in Mexico.

### (1) The problem as a system of linear equations.

The system of equations presented above can be written in matrix form as  $Ax = b$  where

$$x = (I, F, U, O)^\top, \quad b = (0, 0, 0, k)^\top.$$

and the matrix  $A$  is given by

$$A = \begin{pmatrix} -f_{IU} & -f_{FU} & (f_{UI} + f_{UF} + f_{UO}) & -f_{OU} \\ (f_{IU} + f_{IO} + f_{IF}) & -f_{FI} & -f_{UI} & -f_{OI} \\ -f_{IF} & (f_{FU} + f_{FO} + f_{FI}) & -f_{UF} & -f_{OF} \\ 1 & 1 & 1 & 1 \end{pmatrix}.$$

Thus, the steady-state stocks of informal employment, formal employment, unemployment, and out-of-labor-force individuals correspond to the solution of this linear system.

## (2) Contribution of each labor market flows to the variability of the informal employment to population ratio.

Using the gross flow data reported by Leyva and Urrutia (2023), we estimate the contribution of each of the twelve labor market flows described above to the informality in Mexico over time (2005-2021). The variable of interest is defined as

$$y_t = \frac{I_t}{I_t + F_t + U_t + O_t}$$

which represents the proportion of the working-age population that is informally employed.

To assess the contribution of each individual flow, we estimate a separate regression for each transition, including a linear time trend:

$$y_t = \alpha_j + \beta_j f_{AB,t} + \gamma_j t + \varepsilon_{j,t} \quad (5)$$

where  $f_{AB,t}$  denotes the gross flow from state  $A$  to state  $B$  at time  $t$ , with  $A, B \in \{I, F, U, O\}$  and  $A \neq B$ . The coefficient  $\beta_j$  measures the contribution of flow  $AB$  to changes in the informal employment-to-population ratio.

To construct the dependent variable of the regression  $y_t$ , we solve (using the Gauss Jordan method) the linear system described in the previous section, with  $k = 1$ . Under this specification, the solution vector

$$x_t^* = (I_t^*, F_t^*, U_t^*, O_t^*)^\top$$

contains steady-state population shares that sum to one (proportions). Consequently, the

first element of the solution vector,  $I_t^*$ , corresponds directly to the informal employment-to-population ratio, so that  $y_t = I_t^*$ .

For each regression  $j$ , the corresponding design matrix is given by

$$X = \begin{pmatrix} 1 & f_{j,1} & 1 \\ 1 & f_{j,2} & 2 \\ \vdots & \vdots & \vdots \\ 1 & f_{j,n} & n \end{pmatrix},$$

and  $j = 1, 2, \dots, 12$ .

Hence, solving each of the twelve regressions in MATLAB, we obtain the following results:

Table 1: Estimated Contributions of Labor Market Flows to Informality

| Flow     | Estimate | Std. Error |
|----------|----------|------------|
| $B_{IF}$ | 1.3279   | 0.3787     |
| $B_{IU}$ | -0.2884  | 0.5395     |
| $B_{IO}$ | -1.1301  | 0.0903     |
| $B_{FI}$ | 1.2959   | 0.1495     |
| $B_{FU}$ | 0.0467   | 0.8909     |
| $B_{FO}$ | -2.6709  | 0.2810     |
| $B_{UI}$ | 0.7133   | 0.0712     |
| $B_{UF}$ | 0.5334   | 0.0826     |
| $B_{UO}$ | -0.5260  | 0.0471     |
| $B_{OI}$ | 2.1649   | 0.1160     |
| $B_{OF}$ | 1.5635   | 1.1768     |
| $B_{OU}$ | -0.6203  | 0.6144     |

## Interpretation of Results

Each coefficient  $\beta_{AB}$  measures the marginal association between changes in the gross flow from state  $A$  to state  $B$  and changes in the informal employment-to-population ratio. In other words, a positive value of  $\beta_{AB}$  indicates that increases in the corresponding flow are associated with increases in informality, whereas a negative value indicates that increases in the flow are associated with declines in informality.

The largest positive impact on informality corresponds to the transition from out of the labor force to informal employment ( $O \rightarrow I$ ):  $\beta_{OI} = 2.1649$ . This result indicates that the main source of growth in informality is the direct entry of inactive individuals into informal jobs. This suggests that the informal sector operates as an important entry point into the labor market for individuals who were previously outside the labor force.

On the other hand, the largest negative impact on informality corresponds to the transition ( $F \rightarrow O$ ):  $\beta_{FO} = -2.6709$ . This negative value implies that an increase in the number of formal workers exiting into inactivity is associated with a reduction in the informal employment-to-population ratio. An intuitive interpretation is that individuals who leave formal jobs and become inactive do not enter informal employment. As a result, the pool of workers who could potentially transition into informality becomes smaller.

Taken together, these results indicate that informality in Mexico changes mainly because people move in and out of inactivity and informal jobs, not mainly because unemployed people find informal jobs or because workers switch between formal and informal employment.

Regarding statistical precision, several coefficients are estimated with relatively small standard errors, notably those associated with the transitions  $I \rightarrow O$ ,  $F \rightarrow I$ ,  $U \rightarrow I$ , and  $O \rightarrow I$ . This indicates that these flows are robustly associated with changes in the informal employment-to-population ratio.

In contrast, some coefficients display large standard errors, such as those for  $F \rightarrow U$ ,  $O \rightarrow F$ , and  $O \rightarrow U$ , suggesting that the data provide limited information about the contribution of these transitions. Therefore, conclusions should be drawn mainly from the flows whose coefficients are estimated with greater precision.